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Altitude Optimization of Airborne Wind Energy Systems: A Bayesian Optimization Approach

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Abstract— **This study presents a data-driven approach for optimizing the operating altitude of Airborne Wind Energy (AWE) systems to maximize net energy production. Determining the optimal operating altitude of an AWE system is challenging, as the wind speed constantly varies with both time and altitude. Furthermore, without expensive auxiliary equipment, the wind speed is only measurable at the AWE system's operating altitude. The work presented in this paper shows how tools from machine learning can be blended with real-time control to optimize the AWE system's operating altitude efficiently, without the use of auxiliary wind profiling equipment. Specifically, Bayesian Optimization, which is a data-driven technique for finding the optimum of an unknown and expensive-to-evaluate objective function, is applied to the real-time control of an AWE system. The underlying objective function is modeled by a Gaussian Process (GP); then, Bayesian Optimization utilizes the predictive uncertainty information from the GP to decide the best subsequent operating altitude. In the AWE application, conventional Bayesian Optimization is extended to handle the time-varying nature of the wind shear profile (wind speed vs. time). Using real wind data, our method is validated against three baseline approaches. Our simulation results show that the Bayesian Optimization method is successful in dramatically increasing power production over these baselines.**

I. INTRODUCTION

As the fastest growing energy technology for the past decade, wind energy is one of the most promising renewable resources for displacing traditional fossil fuel sources [1]. Airborne Wind Energy (AWE) systems are new paradigm for wind turbines in which the structural elements of conventional wind turbines are replaced with tethers and a lifting body (a kite, rigid wing, or aerostat) to harvest wind power from significantly increased altitudes (typically up to 600m). At those altitudes, winds are stronger and more consistent than ground-level winds.

The vast energy resource from high-altitude winds has attracted the attention of numerous research and commercial ventures over the past two decades [2]–[6]. Besides being able to operate at much higher altitudes than traditional turbines, AWE systems also provide additional control degrees of freedom that allow the systems to *adjust* their operating altitudes and intentionally induce *crosswind motions* to

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Fig. 1. Altaeros Buoyant Airborne Turbine (BAT), Image Credit: [2]

enhance power output. To-date, the majority of research in the area of AWE system control has focused on the latter problem, i.e., crosswind motion control [7]–[11]. Comparatively fewer studies have focused on the impact of adjusting altitude [12]–[14]. However, the first two papers make the assumption of a monotonic wind profile that conforms to a power law model, an assumption that greatly simplifies the altitude optimization problem but is seldom satisfied in the *instantaneous* wind shear (wind speed vs. altitude) profile (as we shall see in this paper). The more recent publication, [14], does take into account the possibility of non-monotonic wind shear profiles but uses an extremum seeking (ES) formulation that only achieves convergence to *local*, rather than global, optima.

Only recently has any effort been undertaken to take into account the *stochastic* nature of the wind shear profile in the context of a *global* altitude optimization. An initial attempt to address this problem using Model Predictive Control (MPC) is detailed in [15]. In the context of the altitude optimization problem, MPC attempts to balance exploration with exploitation in optimizing the altitude setpoint. However, this MPC formulation requires an offline characterization of the statistical properties (conditional mean and conditional variance) of the wind shear (wind speed vs. altitude) profile, thereby necessitating the collection of a very substantial amount of data offline in order to kick-start the optimization.

For the altitude optimization problem at hand, it is desirable to employ a control system that can learn the statistical properties of the wind shear profile online, thereby alleviating the need for offline data collection prior to running the control system. One of the most studied problems in the

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machine learning community is the design of optimization algorithms for a real-world applications using scarce data. In existing literature, this problem has been studied in the context of sequential decision-making problems aiming to learn the behavior of an objective function (called exploration) while simultaneously trying to optimize the objective (called exploitation). As an efficient and systematic approach to balance exploration and exploration in a partially observable environment, Bayesian Optimization has been applied to various real-world problems [16], [17]. In general, Bayesian Optimization aims to find the global optimum of an unknown, expensive-to-evaluate, and black-box function within only a few evaluations. One popular approach is to model the unknown function as a Gaussian Process (GP), where Bayesian Optimization puts prior belief on an objective function to describe the overall structure of that function. At every step of Bayesian Optimization, the next operating point is selected to maximize some *acquisition function*, which characterizes (i) how much will be learned by visiting a candidate point (exploration) and (ii) what the likely performance level will be at that next candidate point (exploitation) [18].

In the case of the altitude optimization problem at hand, the decision variable is the operating altitude at the next time step, denoted by z_{k+1}, where k is the current time step. The objective is net power output, taking under consideration the power required to control the system at a given altitude and the power required to adjust the altitude. In this work, we focus specifically on the Altaeros Energies Buoyant Airborne Turbine (BAT). Fig. 1 shows a picture of this type of AWE system. In fact, the BAT is nominally stationary and hence, it optimizes power by *hunting* for the optimal altitude rather than flying in crosswind patterns.

This paper makes three unique contributions:

- We study the feasibility of employing Bayesian Optimization in the context of altitude optimization for AWE systems.
- We extend the generic version of Bayesian Optimization to capture time-dependent patterns of the wind shear profile. This is done by incorporating *context dependency* into the Bayesian Optimization.
- We validate our proposed algorithms using real wind shear data.

The remainder of this paper is structured as follows: In Section II, we summarize the wind shear profile data and formalize the objective function for the altitude optimization problem. Section III introduces fundamental mathematical formulations, including Gaussian Process (GP) modeling, Bayesian Optimization, and the application of both to the problem at hand. Section IV provides detailed results, demonstrating significant improvement in net energy generation over baseline approaches.

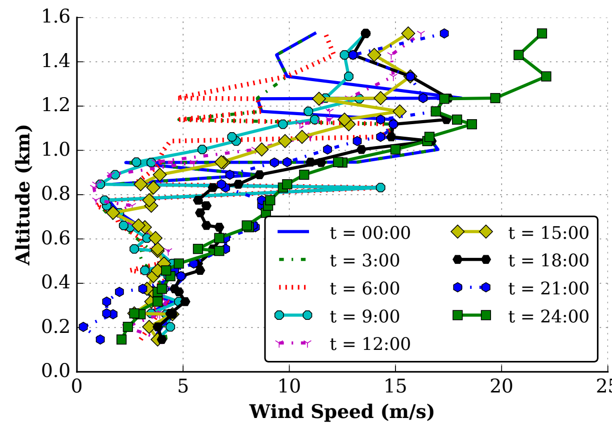


Fig. 2. Wind shear profile based on actual data for one day from Cape Henlopen State Park, Lewes, Delaware, Date: March 1, 2014, [19]. The propagation of wind shear profile at 3 hour increments has been shown. One can conclude that (i) the optimal altitude varies over the course of the day and (ii) the wind shear profile does not fit a neat, monotonic relationship.

II. WIND SHEAR PROFILE, ALTITUDE ADJUSTMENT, AND ENERGY GENERATION MODELS

A. Wind Shear Profile

To maximize generated energy, it is of interest that the AWE system flies at altitudes where the wind velocity, V_{wind}, is as close as possible to the rated wind speed, V_{rated}. Consistent with intuition, if $V_{\text{wind}} < V_{\text{rated}}$, the net power production will diminish because the turbine is producing less power. Less obviously, the net power production will also diminish whenever $V_{\text{wind}} > V_{\text{rated}}$ because more power will be expended controlling the system. While it is possible to install a secondary weather balloon or wind profiler to continuously monitor V_{wind} as a function of altitude (z), such profiling equipment is very costly and still provides no indication of how the wind speed might change at future time instances when the AWE system actually *reaches* its target altitude. Thus, it is of economic interest to use the AWE system itself to periodically explore the wind shear environment, taking into account the spatiotemporal variability of the wind in deciding which altitude to explore next.

To conduct our study, we used data obtained from a Doppler radar-based wind profiler in Cape Henlopen State Park (located in Lewes, Delaware). The data includes wind speed at 30 minute intervals, at 48 altitudes up to $3000m$, over the course of one year [19]. Due to technical constraints for AWE system flight altitude, we used the portion of data involving altitudes up to roughly $1550m$ for the study at hand.

The progression of the wind shear profile over the course of one day, at 3 hour intervals, is shown in Fig. 2. One can infer from this figure that the wind profile at any given time does not follow a power law or logarithmic law (or even a monotonic trend in most instances). Indeed, this is our motivation to introduce a control scheme that is capable of coping with the uncertain nature of the wind shear profile.

B. Altitude Adjustment Model

In any AWE system, a low-level flight control system must exist that regulates key flight variables, such as altitude, pitch angle, and roll angle, to prescribed setpoints. In fact, such a flight control system is described (along with experimental validation results) by one of the co-authors in [6]. Among other things, [6] demonstrates that an underlying flight control system can successfully control the Altaeros BAT to prescribed altitude setpoints, denoted by z_{sp}. Because the efficacy of this lower-level controller has already been demonstrated in the literature, we will work in this paper with a very simple altitude adjustment model that assumes that *the altitude at the next step is equal to the altitude setpoint from the current time step*, i.e.:

$$z_{t+1} = z_t^{\mathrm{sp}}. \tag{1}$$

where z_t^{sp} denotes the altitude setpoint at step t. Given the altitude tracking performance demonstrated in [6] and the sufficiently long time step of 30 minutes that we will work with for the adjustment of z^{sp} in this paper, this very simple model represents a reasonable approximation.

C. Energy Generation Model

In this section, we formalize the objective function of altitude optimization. An appropriate objective function for altitude optimization should account for the energy generated by the turbine and the energy lost by adjusting and maintaining altitude (even when maintaining altitude, control energy must be expended to make small adjustments to the tethers in order to reject typical levels of turbulence). In order to account for each of these objectives, the instantaneous net power generation is expressed as [14]:

$$P_{\mathrm{total}} = c_1 \min\left(V_{\mathrm{wind}}, V_{\mathrm{rated}}\right)^3 - c_2 V_{\mathrm{wind}}^2 + P_z\left(V_{\mathrm{wind}}, \dot{z}\right), \tag{2}$$

In (2), z is the operating altitude, $V_{\mathrm{wind}}(z)$ is the wind speed at the operating altitude, V_{rated} is the turbine's rated wind speed, P_z is the instantaneous power required for or regenerated by adjusting altitude, and $c_{1,2}$ are constant parameters that depend upon the turbine's power curve and flight control system, respectively. Therefore, the first term in (2) characterizes the power generated by the turbine, while the second and third terms represent the control energy required to maintain and adjust altitude, respectively. The power required to adjust altitude, P_z, is given as follows:

$$P_z\left(V_{\mathrm{wind}}, \dot{z}\right) = \begin{cases} \frac{c_3}{\eta_m} \dot{z} V_{\mathrm{wind}}^2 & \dot{z} \leq 0, \\ c_3 \eta_g \dot{z} V_{\mathrm{wind}}^2 & \text{otherwise.} \end{cases} \tag{3}$$

where η_m and η_g represent the motoring and regenerative efficiencies, respectively, of the winches that regulate tether lengths. Moreover, c_3 is a constant parameter that depends on the aerodynamics of the lifting body and its trim pitch angle. Taking into consideration the fact that any upward adjustment of altitude must at some later time be compensated by a downward adjustment (the system must land eventually), we will work in this paper with a simplified version of (2) that lumps motoring and regenerated power into a single term that penalizes instantaneous altitude adjustment [14]:

$$P = c_1 \min\left(V_{\mathrm{wind}}(z), V_{\mathrm{rated}}\right)^3 - c_2 V_{\mathrm{wind}}^2 - \bar{c}_3 V_{\mathrm{wind}}^2 \mid \dot{z} \mid, \tag{4}$$

Here, $\bar{c}_3$ is a lumped parameter that is dependent on overall winch efficiency and aerodynamics of the lifting body. Because the Bayesian Optimization in this paper works in discrete time, using the model in (1), we approximate $\mid \dot{z}_t \mid$ in discrete time by:

$$\mid \dot{z}_t \mid \approx \frac{\mid z_t - z_{t-1} \mid}{\Delta t} \tag{5}$$

where Δt denotes the time step.

Ultimately, equation (4) serves as our objective function, which should be maximized.

III. METHODOLOGY

The ultimate goal of altitude optimization with Bayesian Optimization is to maximize the net energy generated by the BAT. Due to the partially observable nature of the wind shear profile, the optimal altitude is not known *a priori*. Therefore, the optimization strategy must achieve an appropriate balance between learning more about the behavior of the objective function over the control space (exploration) and selecting an altitude that is deemed optimal based on the collected data (exploitation).

A. Bayesian Optimization

Suppose that we want to maximize an unknown, black-box, and expensive-to-evaluate objective function. Due to the cost associated with evaluating this function, it is crucial to select the location of each new evaluation deliberately. For example, in the case of altitude optimization of an AWE system, the task of moving aerostat to new altitude is expensive in terms of time, energy, and cost. Bayesian Optimization represents a promising candidate strategy for this altitude optimization, as Bayesian Optimization focuses specifically on converging to the *global optimum* of an unknown function within *a minimal number of evaluations* on the real system. The major assumption here is that function evaluation is expensive, whereas computational resources are cheap. This matches the altitude optimization problem at hand, where each evaluation of the objective function requires the physical AWE system to relocate itself to a new altitude, necessitating substantial time and energy expenditure.

Broadly speaking, there are two main components to Bayesian Optimization. First, we need an appropriate model of the objective function. A popular approach is to model the underlying function as a Gaussian Process (GP). The selection of a model of the objective function, along with the identification of that function's parameters, can be termed the *learning phase* of Bayesian Optimization. Second, we need to choose an *acquisition function*, which characterizes the fitness of each candidate value of the next operating altitude. The maximization of the expected value of this acquisition function determines the next point (altitude) to operate the system at. The optimization of this acquisition function can be termed the *optimization phase*.

*1) **Learning Phase: Using Gaussian Processes (GPs):*** In this section, we introduce the basic properties of GPs. GPs are used to model the objective function under consideration.

As an attractive choice for non-parametric regression in machine learning, the ultimate goal of GP models is to find an approximation of a nonlinear map, from an input vector to the objective function value. In general, GP models are able to model complex phenomena as they are able to cope with nonlinear effects and interaction between covariates in a systematic fashion. GP models assume that the objective function values associated with different inputs (e.g., altitude), have a joint Gaussian distribution [20].

In general, a GP is fully specified by its mean function, $\mu(z)$, which is assumed to be zero without loss of generality [20] (a coordinate shift can be used to generate a zero mean objective function), and covariance function, $k(z, z')$:

$$P(z) \sim \mathcal{GP}\Big(\mu(z), k(z, z')\Big), \tag{6}$$

In the context of altitude optimization problem, the GP framework is used to predict the generated power, $P(z)$, at candidate altitude, z_c, based on a set of t past observations, $\mathcal{D}_{1:t} = \big\{z_{1:t}, P(z_{1:t})\big\}$.

The power at candidate altitude z_c is modeled to follow a multivariate Gaussian distribution [20]:

$$\begin{bmatrix} y_t \\ P(z_c) \end{bmatrix} \sim \mathcal{N}\Bigg(0, \begin{bmatrix} K_t + \sigma_\epsilon^2 I_t & k_t^T \\ k_t & k(z_c, z_c) \end{bmatrix}\Bigg), \tag{7}$$

where $y_t = \big\{P(z_1), \cdots, P(z_t)\big\}$ is the vector of observed function values. Moreover, vector $k_t(z) = [k(z_c, z_1), \cdots, k(z_c, z_t)]$ encodes the covariances between the candidate altitude, z_c, and the past data points, $z^{1:t}$. The past-data covariance matrix, with entries $[K_t]_{(i,j)} = k(z_i, z_j)$ for $i, j \in \big\{1, \cdots, t\big\}$, characterizes the covariances between pairs of past data points. The identity matrix is represented by I_t and σ_ϵ represents the noise variance [20].

The parametric form for the covariance between two measurement points is prescribed by a *kernel* structure. A common choice for the kernel function, which is used in this work, is a Squared Exponential (SE) kernel, which takes the form:

$$k(z_i, z_j) = \sigma_0^2 \, \exp\Big(-\frac{1}{2}(z_i - z_j)^T \Lambda^{-2}(z_i - z_j)\Big). \tag{8}$$

Qualitatively, the SE kernel structure accounts for the fact that two measurements taken at close altitudes are more likely to be correlated than measurements taken at distant altitudes. The quantitative correlation between different data points is characterized by *hyper-parameters*, which are denoted by $\theta = \big\{\sigma_0, \Lambda\big\}$. Kernel hyper-parameters are identified by maximizing the marginal log-likelihood of the existing observed data, $\mathcal{D}$ [20]:

$$\theta^* = \arg\max_{\theta} \log \, p(y_t \mid z^{1:t}, \theta) \tag{9}$$

Algorithm 1 Bayesian Optimization (BO)

1: **procedure** ALTITUDE OPTIMIZATION WITH BO
2: $\quad \mathcal{D} \leftarrow$ *Initialize*: $\big\{z_{1:2}, P(z_{1:2})\big\}$
3: $\quad$ **for** each iteration **do**
4: $\quad\quad$ Train a GP model from $\mathcal{D}$
5: $\quad\quad$ Compute mean and variance of GP (Eq. (11)-(12))
6: $\quad\quad$ Compute acquisition function (Eq. (14))
7: $\quad\quad$ Find z^* that optimizes acquisition function
8: $\quad\quad$ Append $\big\{z^*, P(z^*)\big\}$ to $\mathcal{D}$

---

where

$$\log \, p(y_t \mid z^{1:t}, \theta) = \Big(-\frac{1}{2}y_t^T K_t^{-1} y_t - \frac{1}{2}\log| \, K_t \, | - \tfrac{t}{2}\log 2\pi\Big). \tag{10}$$

Once the hyper-parameters have been identified, the predictive mean and variance at z_c, conditioned on these past observations, can be expressed as:

$$\mu_t(z_c \mid \mathcal{D}) = k_t(z_c)\Big(K_t + I_t\sigma_\epsilon^2\Big)^{-1} y_t^T, \tag{11}$$

$$\sigma_t^2(z_c \mid \mathcal{D}) = k(z_c, z_c) - k_t(z_c)\Big(K_t + I_t\sigma_\epsilon^2\Big)^{-1} k_t^T(z_c), \tag{12}$$

In summary, the learning phase using GPs involves two main steps: training and prediction. The training step consists of finding proper mean and covariance functions, along with corresponding hyper-parameters, using acquired data (Eq. (9)). The prediction step attempts to characterize the statistical properties of the objective function at the next candidate altitude (Eqs. (11-12)) [20].

*2) **Optimization Phase:*** As mentioned earlier, we need to choose an acquisition function, which is used by Bayesian Optimization to determine the next point to evaluate. Specifically, the acquisition function uses the predictive mean and variance (Eqs. (11-12)) to combine *exploration* of high-variance regions with *exploitation* of high-mean regions.

The ultimate goal of Bayesian Optimization is to determine the next point based on past observations. Among several choices of acquisition functions, we use an acquisition function belonging to the improvement-based family [18]. More precisely, the next operating point is selected as the one that maximizes the so-called expected improvement:

$$z_{t+1} = \arg\max_{z} \, \mathbb{E}\Big(\max\big\{0, P_{t+1}(z) - P(z)^{max}\big\} \mid \mathcal{D}_{1:t}\Big) \tag{13}$$

where $\max\big\{0, P_{t+1}(z) - P(z)^{max}\big\}$ represents the *improvement* toward the best value of the objective function so far, $P(z)^{max}$. This quantity is referred to as *expected improvement* and will be denoted by $\mathbb{EI}$. Expected improvement can be expressed in closed form as [21]:

$$\mathbb{EI}(z_{t+1} \mid \mathcal{D}_{1:t}) = \begin{cases} \Big(\mu_t(z) - P(z)^{max}\Big)\Phi(Z) + \sigma_t(z)\phi(Z), & \sigma_t(z) > 0 \\ 0, & \sigma_t(z) = 0 \end{cases} \tag{14}$$

where

$$Z = \frac{\mu_t(z) - P(z)^{max}}{\sigma_t(z)}, \tag{15}$$

and $\Phi(.)$ and $\phi(.)$ denote the cumulative and probability density function for the normal distribution, respectively.

We summarize the conventional Bayesian Optimization in Algorithm 1. The algorithm is initialized by two previously-evaluated altitude and corresponding function values (line 2). Then, at each step, a GP model is trained (line 4) to compute the predictive mean and variance (line 5). These statistical quantities are used to construct the acquisition function (line 6). Next, the point that maximizes the acquisition function is selected as the next altitude (line 7). At the next optimization instance, this point is added to the historical data (line 8), and the process repeats.

B. Context-Dependent Optimization

The time-varying nature of our objective function (the wind speed at a given altitude does not remain constant over time) compels us to modify the conventional Bayesian Optimization method (Algorithm 1) to account for *spatiotemporal* variation of the wind shear profile. One way to cope with the dynamic nature of objective function is through contextual GP [22]. More precisely, one of the benefits of modeling the objective function using GPs is that we can characterize the statistical properties of the system output based on environmental variables, called *context*. To deal with time-varying nature of our objective function, we consider *time* as the context.

Employing contextual GP is very straightforward. In Section III-A.1, the kernel of the GP is defined only in terms of altitude. To account for time-dependency, we modify the kernel structure such that it consists of an addition of two kernels, one over altitude, k_z, and one over time, k_t:

$$k\Big((z,t),(z',t')\Big) = k_z(z,z') + k_t(t,t'), \tag{16}$$

where

$$k_z(z,z') = \sigma_{0z}^2 \, \exp\Big(-\frac{1}{2}(z-z')^T\Lambda_z^{-2}(z-z')\Big) \tag{17}$$

and

$$k_t(t,t') = \sigma_{0t}^2 \, \exp\Big(-\frac{1}{2}(t-t')^T\Lambda_t^{-2}(t-t')\Big) \tag{18}$$

In Eq. (17) and (18), $\theta_z = \{\sigma_{0z}, \Lambda_z\}$ and $\theta_t = \{\sigma_{0t}, \Lambda_t\}$ denote the hyper-parameters of k_z and k_t, respectively.

Context-Dependent Bayesian Optimization is summarized in Algorithm 2. First, at the very first two time steps, the procedure is initialized by two previously evaluated altitudes and the corresponding function values (line 2). Then, at each time step, the context (time) is fixed at the current instance (line 4). Next, the composite kernel is constructed based upon Eq. (16) (line 5). Similar to Algorithm 1, the GP model and its predictive mean and variance are computed (line 6-7). Then, the acquisition function value is computed by Eq. (14), and optimization is done only over the altitude space (line 8-9). Finally, the suggested candidate (altitude) and the current time are augmented to the data (line 10) and the process is restarted.

Algorithm 2 Context-Dependent Bayesian Optimization (CDBO)

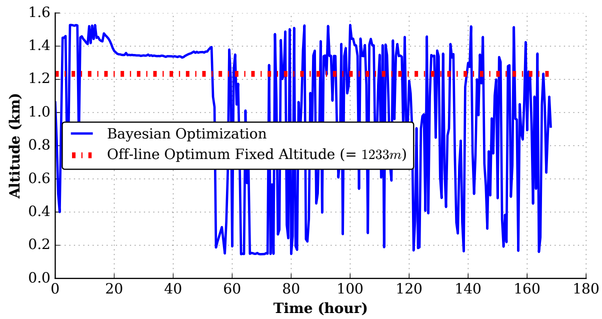
```
1:  procedure ALTITUDE OPTIMIZATION WITH CDBO
2:      D ← Initialize: {(z_{1:2}, t_{1:2}), P(z_{1:2})}
3:      for each time step do
4:          Set the context into the current time
5:          Construct a composite kernel (Eq. (16))
6:          Train a GP model from D
7:          Compute predictive mean and variance of GP
8:          Compute acquisition function
9:          Find z* that optimizes acquisition function
10:         Append {(z*, t_{current time}), P(z*)} to D
```

IV. RESULTS

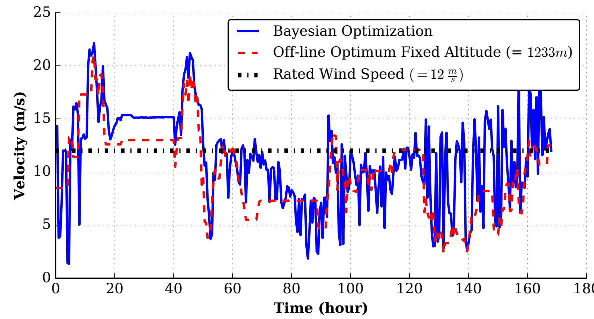
We evaluated the proposed altitude optimization method by simulating the model in Sec. II based on available data, acquired by a wind profiler in Cape Henlopen, DE [19]. Here, we compare the system performance under four different control scenarios:

- **Scenario 1: Off-line Optimum Fixed Altitude:** The first scenario involves flying the AWE at an *optimum fixed altitude*. We identify this optimal fixed altitude by off-line calculation of energy generation in different altitudes using given wind data. It is important to note that this calculation requires knowledge of the future. Hence, this scenario serves an upper bound on the potential of any fixed altitude algorithm but does not constitute an implementable real-time control strategy.
- **Scenario 2: Minimum Fixed Altitude:** The second scenario is representative of conventional towered wind turbines. For this scenario, we simulate the performance of the system operating at the lowest available altitude ($146m$), which is comparable to the hub height of the world's tallest towered wind turbines.
- **Scenario 3: Maximum Fixed Altitude:** The third scenario involves flying the AWE at the maximum tabulated fixed altitude, $1528m$.
- **Scenario 4: Bayesian Optimization:** The fourth scenario employs Bayesian Optimization, as outlined in Algorithm 2.

The resulting operating altitude and corresponding wind velocity are shown in Fig. 3. Furthermore, Fig. 4 shows the net energy production across all scenarios. One can conclude from this figure that the Bayesian Optimization method



(a) Operating altitude vs. time for Bayesian Optimization (compared with the offline-optimized constant altitude)



(b) Comparison of wind speeds at operating altitude

Fig. 3. Comparison of operating altitude and wind velocity for different control strategies over the course of one week.

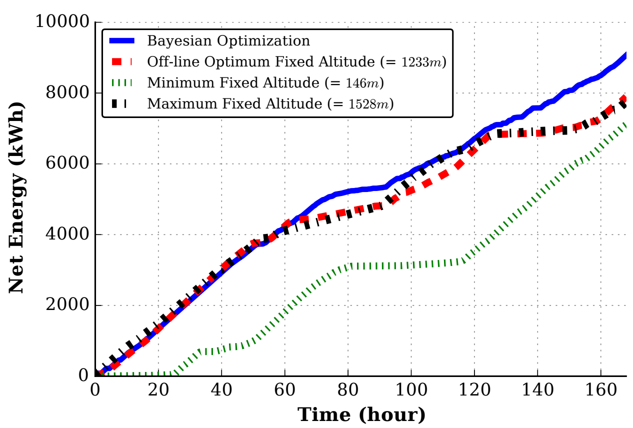


Fig. 4. Comparison of net energy production over the course of one week. Bayesian Optimization outperforms off-line optimum fixed altitude, minimum fixed altitude, and maximum fixed altitude.

results in superior net energy production when compared to other scenarios.

V. Conclusions

We presented a data-driven approach for optimizing the altitude of an AWE system to maximize the system's net energy output. In implementing this altitude optimization approach, we extended conventional Bayesian Optimization to capture time-dependent patterns of our objective function. For a model of the Altaeros BAT, we showed that Bayesian Optimization improves the resulting net energy generation, as compared to other control scenarios.

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